Survival Analysis

I collected my thoughts on Survival Analysis, with examples, proofs and questions I want to return to as I learn more.

cox model

[@cox1972]cox # 1. Regression Models

1.1 One Individual's Data

For an individual with covariates $z_1, \ldots, z_p$, the failure time is represented as t_i. The failure rate, also known as the hazard function, for this individual is given by:

$$\lambda(t \mid z_i) = \exp(z_i^T \beta) \times \lambda_0(t)$$

Where: - z_i: vector of covariates for individual i, - β: coefficient vector that needs to be estimated, - $\lambda_0(t)$: baseline hazard function (i.e., the hazard when $z_i = 0$).

⟵ **Proof idea** Check the initial and boundary conditions after finding a general solution.

1.2 Proportional Hazard Assumption

Under the proportional hazard assumption, the ratio of hazard functions between two individuals with covariates z_i and z_j remains constant over time:

$$\frac{\lambda(t \mid z_i)}{\lambda(t \mid z_j)} = \exp((z_i - z_j)^T \beta)$$

This property is what makes the Cox proportional hazards model useful in survival analysis.

1.3 Likelihood Function

For censored data, the likelihood function of the hazard rate for n individuals is given by:

$$L(\beta) = \prod_{i=1}^{n} [\lambda(t_i \mid z_i)]^{\delta_i} \exp\left(-\int_0^{t_i} \lambda(s \mid z_i) ds\right)$$

⟵ **My thought** Inference uses data to distinguish plausible parameters; assumptions still matter.

Where: - t_i: observed failure or censoring time for individual i, - δ_i: event indicator for individual i, where $\delta_i = 1$ if the event is observed (failure) and $\delta_i = 0$ if censored.

For computational ease, the partial likelihood function, $L_P(\beta)$, can be expressed as:

$$L_P(\beta) = \prod_{i=1}^{n} \frac{\exp(z_i^T \beta)}{\sum_{j \in R(t_i)} \exp(z_j^T \beta)}$$

Where: - $R(t_i)$: the set of individuals still at risk at time t_i.

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2. Product-Limit Method for Censored Data (Kaplan-Meier Estimator)

⟵ **Motivation** What does the definition really control? Try to draw it before proving it.

The Kaplan-Meier estimator is used to estimate the survival probability in the presence of censored data. For n independent individuals, the survival probability is given by:

$$S(t) = P(T \geq t) = \prod_{j:t_j \leq t} \left(1 - \frac{d_j}{Y_j}\right)$$

Where: - d_j: the number of failures at time t_j, - Y_j: the number of individuals at risk just before time t_j.

This estimator assumes that censored data are randomly distributed.

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3. Log-Rank Test (2-Sample Comparison)

⟵ **Proof idea** Identify the exact statement before proving it.

3.1 Overview

The log-rank test is used to compare the survival distributions of two groups. It is a non-parametric test that assumes that the hazard functions are proportional.

- **Slight advantage**: Is the difference significant enough? This is determined by calculating the test statistic.
- **At first glance**: This can be thought of as a generalization of the t-test to survival data, where the censoring mechanism is taken into account.

1-sample Kaplan-Meire

Unfolds in time

conditional probability

handel more on censoring data(into conditional probability)

with standard error–Green wood's formula

3.2 Handling Ties (Log-Rank Test Table)

⟵ **My thought** The theorem is easier to remember when its hypothesis feels necessary.

	Group 1	Group 2	Total
Died	d_{1k}	d_{2k}	d_k
Survived			
Total	r_{1k}	r_{2k}	r_k

Where: - d_k: total number of deaths at time t_k, - r_k: number of individuals at risk at time t_k, - d_{1k}, d_{2k}: number of deaths in group 1 and group 2 at time t_k, respectively.

3.3 Test Statistic

To test the null hypothesis H_0: No difference between survival curves (i.e., $S_1(t) = S_2(t)$), the test statistic is computed as:

$$W = \frac{\sum_{k=1}^{K} (O_k - E_k)}{\sqrt{\text{Var}(O_k)}}$$

Where: - O_k: observed number of events in group 1 at time t_k, - E_k: expected number of events under the null hypothesis, - $\text{Var}(O_k)$: the variance of O_k.

The expectation and variance are calculated as:

$$E(O_k) = \frac{r_{1k}}{r_k} d_k$$

$$\text{Var}(O_k) = \frac{r_{1k} r_{2k} d_k (r_k - d_k)}{r_k^2 (r_k - 1)}$$

Under the null hypothesis:

$$W \sim N(0, 1)$$

The p-value can be computed from this statistic to assess the significance.

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4. Cox (1972) Proportional Hazards Model

⟵ **Motivation** Decomposing a complicated signal into simpler modes makes dynamics easier to understand.

4.1 Failure Time

Let T be a continuous random variable representing the failure time. The survival function is defined as:

$$S(t) = P(T \geq t) = 1 - F(t)$$

Where $F(t)$ is the cumulative distribution function (CDF) of T.

4.2 Discrete Case

In the discrete case, the hazard function $\lambda(t_k)$ at time t_k is defined as:

$$\lambda(t_k) = \frac{P(t_k \leq T < t_{k+1} \mid T \geq t_k)}{h_k} = \frac{f(t_k)}{S(t_k)}$$

Where $f(t_k)$ is the probability density function (PDF) of T at time t_k.

4.3 Continuous Case

⟵ **Proof idea** Begin with epsilon-delta or a sequence characterization.

In the continuous case, the hazard function $\lambda(t)$ is defined as:

$$\lambda(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} = \frac{f(t)}{S(t)}$$

The hazard function represents the instantaneous failure rate at time t given survival until time t.

The relationship between the hazard function and the survival function is:

$$\lambda(t) = -\frac{d}{dt} \log S(t)$$

Which implies that:

$$S(t) = e^{-\int_0^t \lambda(u) du}$$

4.4 Likelihood Function for Censored Data

For censored data, the likelihood function can be expressed as:

$$L = \prod_{i=1}^{n} [\lambda(t_i)]^{\delta_i} [S(t_i)]^{1-\delta_i}$$

Where: - $\delta_i = 1$ if the event is observed (failure), - $\delta_i = 0$ if censored.

This likelihood can be maximized to estimate the coefficients β in the Cox model.

4.5 log rank test for cox

References of relavent paper

⟵ **My thought**
Censoring changes the information available, not merely the sample size.

Survival Analysis Study Notes

From Cox regression to residual diagnostics, time-dependent effects, AFT models, competing risks and frailty.

I have been trying to connect these topics rather than remembering them as separate models. For me, the main question is always: **what is being modeled—instantaneous hazard, the event-time distribution, or the probability of a particular type of event?** The notation is similar, but their interpretations are not interchangeable.

Notation I will use throughout

Let T denote event time, C a censoring time, $t_i = \min(T_i, C_i)$ the observed time, $\delta_i = \mathbf{1}(T_i \le C_i)$ the event indicator, and x_i a covariate vector. In the Cox sections, $h(t \mid x)$ denotes the hazard and $h_0(t)$ the baseline hazard. I write

$$S(t \mid x) = P(T > t \mid x), \qquad H(t \mid x) = \int_0^t h(u \mid x)\, du, \qquad S(t \mid x) = e^{-H(t|x)}.$$

At distinct event times $t_{(1)} < \cdots < t_{(d)}$, R_i is the set still at risk immediately before $t_{(i)}$. A risk score is $\eta_i = x_i^\top \beta$, and a hazard ratio compares two hazards at the **same** instant.

← **My thought**

My map: Cox studies relative hazards; AFT studies the time scale; competing-risks models ask which event happens.

Cox proportional hazards and partial likelihood

Why a semi-parametric model?

In clinical trials and observational medical studies, subjects differ in treatment, age, disease state and many other characteristics. Kaplan−Meier curves and log-rank tests help compare groups, but a regression model is needed when I want several risk factors in the same analysis. A Cox model describes the influence of the covariates without forcing a particular distribution on the baseline time-to-event pattern. That is the semi-parametric idea: the coefficient vector β is finite-dimensional, while the function $h_0(t)$ is left unspecified.

The Cox proportional-hazards model is

$$\boxed{h(t \mid x) = h_0(t) \exp(x^\top \beta)}.$$

For the baseline covariate vector $x = 0$, the hazard is $h_0(t)$. When zero is not an interesting reference level (for example, an age of zero), covariate centering can give the baseline a more meaningful interpretation.

A particularly important structural point: a constant **hazard ratio** does not require constant hazards. Both individuals' hazards may rise or fall as time passes. They change together by a fixed multiplicative factor. A model with *any* time-varying risk ratio is not necessarily a proportional-hazards Cox model.

Relative risk and how I interpret coefficients

For two subjects with vectors x_a, x_b,

$$\frac{h(t \mid x_a)}{h(t \mid x_b)} = \exp\{(x_a - x_b)^\top \beta\},$$

which is independent of t. This explains why a one-unit increase in x_j, holding the other predictors fixed, multiplies hazard by e^{β_j}.

- $\beta_j > 0$: the hazard ratio is above one; the covariate is associated with a higher event hazard.
- $\beta_j = 0$: the model assigns no change in hazard to a one-unit difference.
- $\beta_j < 0$: the hazard ratio is below one; the event hazard is lower.

For a binary treatment indicator $x \in \{0, 1\}$, the treated-to-control hazard ratio is e^{β}. For a continuous predictor, the ratio for a difference Δx is $e^{\beta \Delta x}$, not necessarily e^{β} unless the change is exactly one unit. In a multivariable model, an association is *conditional on the other included predictors*.

Simple treatment example. If $x = 1$ means a new cancer treatment and $x = 0$ the standard treatment, then $e^{\beta} < 1$ means the new group has a lower hazard at each time, assuming proportional hazards. It does **not** mean its survival time is multiplied by $1/e^{\beta}$; that is a different, AFT-style interpretation.

⟵ **My thought**

Why no intercept? A constant β_0 can be absorbed into $h_0(t)$; it is not separately identified.

Survival curves under the Cox model

The accumulated baseline hazard is $H_0(t) = \int_0^t h_0(u)\, du$. For time-independent x,

$$H(t \mid x) = H_0(t) e^{x^\top \beta}, \qquad \boxed{S(t \mid x) = \{S_0(t)\}^{e^{x^\top \beta}}}.$$

I find this identity helpful because it connects hazard-scale regression to an entire survival curve. In a correctly specified model with fixed covariates, survival curves for subjects whose risk scores are ordered cannot cross: applying two different positive powers to the same $S_0(t) \in (0, 1)$ preserves their ordering. This is a **diagnostic intuition**, not a proof that observed KM estimates can never cross by random sampling fluctuation.

Taking logarithms twice gives

$$\log\{-\log S(t \mid x)\} = \log H_0(t) + x^\top \beta.$$

Thus, plots of $\log[-\log \widehat{S}(t)]$ against $\log t$ for categories of a fixed covariate should be approximately parallel when the proportional-hazards relation is plausible. Substantial crossing of Kaplan – Meier curves is another reason to question proportional hazards.

Where partial likelihood comes from

The clever part of Cox regression, in my view, is that we can learn β without first specifying $h_0(t)$. Suppose failures occur at distinct times, censoring is suitably non-informative, and i is the subject failing at time $t_{(i)}$. Conditional on one event occurring among the risk set R_i, the approximate chance that it is subject i is

$$P(i \text{ fails} \mid \text{one event in } R_i) = \frac{h_0(t_{(i)}) e^{x_i^\top \beta}}{\sum_{j \in R_i} h_0(t_{(i)}) e^{x_j^\top \beta}} = \frac{e^{x_i^\top \beta}}{\sum_{j \in R_i} e^{x_j^\top \beta}}.$$

The same baseline hazard appears in the numerator and denominator and cancels. Multiplying those conditional terms over observed failures gives the **partial likelihood**:

$$\boxed{L_p(\beta) = \prod_{i:\delta_i = 1} \frac{\exp(x_i^\top \beta)}{\sum_{j \in R_i} \exp(x_j^\top \beta)}}.$$

The product is over failures, *not* over censored observations as new numerator terms. Censored individuals may nevertheless contribute to denominators before leaving their risk sets. The formula above assumes no tied failures; ties require an appropriate treatment, such as Breslow or Efron approximation.

I should distinguish this from the complete likelihood of (T_i, δ_i), which also involves the baseline hazard. Partial likelihood uses information in **which subject fails among those at risk**.

Score, weighted mean and information matrix

The partial log-likelihood is

$$\ell_p(\beta) = \sum_{i:\delta_i=1} \left[x_i^\top \beta - \log \left\{ \sum_{j \in R_i} e^{x_j^\top \beta} \right\} \right].$$

Define the risk-set weights and weighted mean

$$w_{ij}(\beta) = \frac{e^{x_j^\top \beta}}{\sum_{k \in R_i} e^{x_k^\top \beta}}, \qquad \bar{x}_i(\beta) = \sum_{j \in R_i} w_{ij}(\beta) x_j.$$

Differentiating gives the score function

$$U(\beta) = \frac{\partial \ell_p}{\partial \beta} = \sum_{i:\delta_i=1} \{x_i - \bar{x}_i(\beta)\}.$$

The idea is lovely: the score compares the covariate of the person who failed to the risk-weighted mean among those who could have failed. The observed information is a sum of risk-weighted covariance matrices,

$$I(\beta) = -\frac{\partial^2 \ell_p}{\partial \beta \partial \beta^\top} = \sum_{i:\delta_i=1} \sum_{j \in R_i} w_{ij}(\beta) \{x_j - \bar{x}_i(\beta)\}\{x_j - \bar{x}_i(\beta)\}^\top.$$

A numerical optimizer, often Newton–Raphson, solves $U(\hat{\beta}) = 0$. Under the usual large-sample regularity conditions,

$$\widehat{\mathrm{Var}}(\hat{\beta}) = I(\hat{\beta})^{-1}.$$

⟵ **My thought**

Proof idea: differentiate log-sum-exp. The first derivative is a weighted mean; the second derivative is a weighted covariance.

Wald tests, confidence intervals and hazard ratios

For a single parameter, the approximate Wald statistic is

$$Z_j = \frac{\hat{\beta}_j}{\mathrm{se}(\hat{\beta}_j)} \overset{H_0:\beta_j=0}{\approx} N(0,1).$$

An approximate 95% confidence interval on the coefficient scale is $\hat{\beta}_j \pm 1.96\,\mathrm{se}(\hat{\beta}_j)$. A confidence interval for the hazard ratio is obtained by exponentiating **both endpoints**:

$$\mathrm{CI}_{95\%}(HR_j) = \left[e^{\widehat{\beta}_j - 1.96\,\mathrm{se}(\widehat{\beta}_j)}, e^{\widehat{\beta}_j + 1.96\,\mathrm{se}(\widehat{\beta}_j)} \right].$$

The delta method also gives the approximate standard error of $e^{\widehat{\beta}_j}$ as $e^{\widehat{\beta}_j}\mathrm{se}(\hat{\beta}_j)$. Exponentiating coefficient confidence limits is usually easier to interpret and respects positivity. A large-sample approximation does not assert that every finite-sample estimator is exactly unbiased.

Working interpretation for daily relapse. The clinic 2 indicator has a reported coefficient -1.0091 and hazard ratio 0.36 (95% CI 0.24 to 0.55); a prison-record indicator has HR 1.37 (CI 0.98 to 1.90); dose has HR about 0.96 per unit (CI 0.95 to 0.98). I would say that clinic group and dose show evidence of association on the reported scale, while prison record is less conclusive under a conventional two-sided 5% criterion. These are **associations within the fitted model** and need the study design for causal interpretation.

For surgery and radiotherapy indicators, the fitted relative hazard for having both rather than neither is $\exp(\beta_{\text{surg}} + \beta_{\text{rad}})$ if the model includes their main effects but no interaction.

Estimating the baseline hazard and baseline survival

After obtaining $\hat{\beta}$, the baseline hazard still needs estimation. For d_i events at time $t_{(i)}$, a Breslow / Nelson–Aalen-style increment is:

$$\Delta\widehat{H}_0(t_{(i)}) = \frac{d_i}{\sum_{j \in R_i} \exp(x_j^\top \hat{\beta})}, \qquad \widehat{H}_0(t) = \sum_{t_{(i)} \le t} \Delta\widehat{H}_0(t_{(i)}).$$

Then one commonly uses $\widehat{S}_0(t) = \exp\{-\widehat{H}_0(t)\}$; a product-limit-type baseline survival estimator can also be derived by scaling conditional failure probabilities. Any individual's fitted survival curve follows by

$$\widehat{S}(t \mid x) = \{\widehat{S}_0(t)\}^{\exp(x^\top \widehat{\beta})}.$$

The two-stage nature of Cox modeling is what I want to remember: estimate relative risks without specifying baseline shape, then estimate baseline if absolute survival predictions are required.

Design requirements and limits of interpretation

Sound design matters before fitting a complex regression model: consider sample representativeness, event counts relative to the number of parameters (an illustrative *10 events per parameter* heuristic), an excessive censoring fraction, imbalanced events across groups, and careful predictor coding. I should treat an events-per-variable rule as a planning heuristic rather than a universal guarantee of validity.

A nominal variable with four categories, such as blood type, needs indicator variables relative to a reference group—not a single numeric $1, 2, 3, 4$ score unless an actual ordered linear effect is defensible. Continuous-variable transformations should be motivated by substantive interpretation and diagnostic evidence, not chosen simply to manufacture significance. If proportional hazards is doubtful, later topics introduce residual tests, time-dependent effects and stratified models.

My topic-4 review questions: Where does the baseline cancel? Why do censored people still enter risk sets? How do I get a confidence interval for HR? What changes when hazards are not proportional?

Goodness of fit and predictive performance

Four things I want to check after fitting a Cox model

Model adequacy involves proportional-hazards assumptions, accuracy of the predicted event pattern, influential or unusual subjects, and whether the covariates have a sensible functional form. In linear regression I might inspect ordinary residuals, R^2 and an F test. Censoring and a time-to-event outcome require a different toolkit: partial-likelihood comparisons, Cox−Snell / martingale / Schoenfeld residuals, concordance, and time-specific discrimination.

Likelihood ratio test for nested Cox models

Let $M_0 \subset M_1$ be nested models with maximized partial likelihoods L_0, L_1. If M_1 adds q parameters, the likelihood ratio statistic is

$$\Lambda = -2 \log \frac{L_0}{L_1} = 2\{\ell_1(\hat{\beta}_1) - \ell_0(\hat{\beta}_0)\}.$$

Under the null and appropriate large-sample conditions, Λ has an approximate χ^2_q distribution. Larger values mean stronger evidence for the additional terms; the p value is $P(\chi^2_q \ge \Lambda)$. This overall comparison differs from individual Wald tests. For example, a treatment-only Cox model can be compared with

a treatment-plus-mutation-status model; whether the mutation predictor improves fit is a nested-model question.

Pseudo-R^2 and why it is not ordinary regression R^2

Explained-variation measures can be defined by analogy with R^2. One likelihood-based version is

$$R^2_M = 1 - \left(\frac{L_0}{L_M}\right)^{2/n},$$

where L_0 is the baseline/null partial likelihood, L_M the fitted-model partial likelihood, and n the sample size. Unlike ordinary least-squares R^2, there is no simple decomposition of total event-time variance into fitted and residual sums of squares. Several pseudo-R^2 definitions coexist; I need to name the one I report and avoid calling it a literal percentage of explained survival-time variation.

Concordance index: ranking predicted risk

The risk score is $\eta_i = x_i^\top \hat{\beta}$. Higher η_i predicts higher hazard and, in the model, earlier events in a stochastic sense. For two observed uncensored times, the pair is concordant if

$$(T_i - T_j)(\hat{\eta}_i - \hat{\eta}_j) < 0.$$

The **c-index** is the proportion of comparable ordered pairs whose risk-score order matches the observed survival ordering, with conventions for ties. A value near 0.5 represents little ranking information, while a value toward one represents stronger concordance. Censoring determines which pairs can be compared, so I should not simply calculate a naive rank correlation on all observed times.

A model can rank patients reasonably well and still produce poorly calibrated absolute event probabilities. Thus **discrimination** and **calibration** are different tasks.

ROC, AUC, Youden index and time-dependent ROC

For a binary outcome, an ROC curve plots sensitivity against $1-$specificity as the cutoff varies; the area under it (AUC) summarizes discriminatory ability. The conventional ROC x-axis is the **false positive rate**, $1 -$ specificity. AUC 0.5 corresponds to chance discrimination and 1 to perfect separation. The **Youden index** for a chosen threshold is

$$J = \text{sensitivity} + \text{specificity} - 1,$$

and the maximum J is one possible way to select a threshold, if the clinical consequences of false positives and false negatives are suitably considered.

With survival times, whether a person is a "case"depends on time. It is useful to distinguish:

Cumulative/dynamic definition at time t:

$$\text{Se}_t(c) = P(\eta > c \mid T \le t), \qquad \text{Sp}_t(c) = P(\eta \le c \mid T > t).$$

Incident/dynamic definition: the cases are failures occurring around time t, while controls remain event-free beyond t. Estimating either ROC/AUC curve correctly requires accounting for censoring. Reporting an unqualified AUC without a time horizon may hide an essential part of what was predicted.

⟵ **My thought**

My thought: a high c-index is not the same as a reliable individual survival probability. Prediction requires both discrimination and calibration.

Overfitting, underfitting and validation

Overfitting: very good apparent performance on training observations, much worse performance on new observations. **Underfitting:** both training and external/test fit are poor because the model misses important structure. Common validation approaches include train/test splitting (illustrative proportions from 70:30 to 90:10) and K-fold cross-validation (10 folds as a common choice; more folds also

discussed). The principle matters more than any fixed split: model selection and hyperparameter tuning should happen inside resampling, not after inspecting the test results.

For survival models, validation must also respect censoring and the intended prediction horizon. I would report a discrimination measure, an appropriate calibration assessment, and uncertainty/validation procedure rather than choosing a model on its training-data p values alone.

Cox model checking using residuals

Why ordinary residuals are not enough

A censored observation does not reveal the true event time. Consequently, $T_i - \widehat{T}_i$ cannot play the same role as a residual in linear regression. Three complementary diagnostic quantities are: **Cox–Snell residuals** for overall survival-distribution adequacy, **martingale residuals** for functional form and unusual observations, and **Schoenfeld residuals** for the proportional-hazards assumption.

Cox–Snell residuals and the exponential transformation

Under a correctly specified continuous survival model, the cumulative hazard evaluated at the true event time has a standard exponential distribution:

$$H(T \mid x) = -\log S(T \mid x) \sim \text{Exp}(1).$$

For an observed time t_i, the fitted Cox–Snell residual is

$$\hat{r}_{Ci} = \widehat{H}_0(t_i) \exp(x_i^\top \hat{\beta}),$$

with $\widehat{H}_0$ constructed from the fitted risk sets. Under adequate specification, an empirical cumulative hazard calculated from the residuals, with censoring retained, should be close to $y = r$. This is the familiar **45-degree-line** diagnostic. Marked systematic departure suggests poor overall fit, but it does not by itself identify which assumption failed.

If $\Delta\widehat{H}_0(t_j) = d_j / \sum_{k \in R_j} \exp(x_k^\top \hat{\beta})$, then I can compute each $\hat{r}_{Ci}$ without having to choose an arbitrary parametric form for h_0. A hospital readmission example compares two specifications; in words, the better-fitting one is the model whose residual cumulative-hazard pattern stays closer to the reference line.

Martingale residuals: observed minus expected events

The residual

$$\boxed{\widehat{M}_i = \delta_i - \hat{r}_{Ci}}$$

is the observed event count (zero or one) minus the fitted cumulative event intensity. It lies in $(-\infty, 1]$, so its distribution need not be symmetric. A large positive residual can reflect a failure occurring despite a relatively small predicted cumulative hazard; a large negative residual can arise when predicted cumulative hazard is high but the event has not been observed.

Why "martingale"? Let $N_i(t)$ count whether person i has failed by time t, and let $Y_i(t)$ indicate that person is under observation and at risk just before t. If the intensity is $Y_i(t)h(t \mid x_i)$, the compensated process

$$M_i(t) = N_i(t) - \int_0^t Y_i(u)h(u \mid x_i)\, du$$

has the martingale property under the statistical model: conditional on the observed past, it has no future expected drift. The integral is the **compensator**. This gives a mathematical reason for thinking about residuals as observed minus expected counts rather than event time minus predicted time.

Looking for the correct functional form

Suppose a continuous predictor is currently omitted or modeled linearly. Plotting fitted martingale residuals against that predictor and examining a smooth trend can suggest whether a transformation such as $\log x$, x^2 or a flexible nonlinear function might be helpful. A marked systematic trend is a reason to revisit the model; a largely patternless cloud is consistent with the current functional specification. When the residuals are plotted against a predictor already in the model, interpretation should be sensitive to the fitting and smoothing procedure.

Martingale methods also lie behind the asymptotic theory of KM estimation, log-rank testing and Cox partial likelihood—not merely a diagnostic trick.

⟵ **My thought**

Proof idea: $N_i(t)$ counts what occurred; the integral of intensity predicts how many events were expected. Their difference is the martingale residual.

Schoenfeld residuals: asking whether effects change with time

At an uncensored event time $t_{(i)}$, the Schoenfeld residual for the covariate vector is the observed event covariate minus its risk-weighted mean:

$$\boxed{r_{Si} = x_i - \frac{\sum_{j \in R_i} x_j \exp(x_j^\top \widehat{\beta})}{\sum_{j \in R_i} \exp(x_j^\top \widehat{\beta})}}.$$

Each event contributes a residual for each predictor; censored observations do not directly supply their own Schoenfeld event residual. A graph of appropriately **scaled Schoenfeld residuals versus time**, often with a smooth trend, helps detect time variation in a regression coefficient. Under proportional hazards, the coefficient should remain approximately constant; a systematic time trend is evidence against that assumption.

For example, with drug dose x_1 and age x_2, each event produces $r_{Si,1}$ and $r_{Si,2}$. I would inspect their time trends separately as well as use a global proportional-hazards test.

Diagnostic comparison and alternatives

Diagnostic	Construction	Main question
Cox–Snell	Fitted cumulative hazard of observed time	Does the overall fitted event-time pattern make sense?
Martingale	$\delta_i - \widehat{H}_i(t_i)$	Is the covariate function sensible? Any unusual event pattern?
Schoenfeld	Observed covariate minus risk-set mean at event	Does a fitted coefficient vary over time?

If proportional hazards fails, consider time-varying effects, stratification, or alternative time-to-event regression structures rather than forcing the original Cox model. One practical illustration is splitting a smoker's follow-up at the time of smoking cessation, so that smoking status changes between intervals. This leads naturally to time-dependent covariates.

Time-dependent covariates and stratified Cox models

Two meanings of "time-dependent"that I must not confuse

A **time-varying covariate with a constant effect** has a current value $x(t)$ and coefficient β:

$$h(t) = h_0(t)\exp\{\beta x(t)\}.$$

For example, CD4 count or blood pressure can be repeatedly updated. At any given time, a one-unit difference in the current covariate still implies $HR = e^{\beta}$. A **time-varying effect of a fixed covariate** instead uses $\beta(t)$:

$$h(t) = h_0(t)\exp\{\beta(t)x\},$$

so the hazard ratio for a fixed one-unit difference is $\exp\{\beta(t)\}$ and changes with time. One example is $\beta(t) = \beta_1 + \beta_2 t$, equivalently adding the interaction xt as another covariate.

Time-varying covariates can be **internal**, relating to the evolving state of the person (blood counts, disease progression), or **external**, with an exogenous or predetermined trajectory (air temperature, prespecified drug dosing). Their use for prediction differs because an internal predictor's *future* values may not yet be known.

Data structure and risk-set updating

The data for subject i consist of $(t_i, \delta_i, \{x_i(u) : 0 \le u < t_i\})$. The current covariate values just before each failure time determine risk-set contributions. The extended Cox model is

$$\boxed{h_i(t) = h_0(t)\exp\left\{\sum_{k=1}^{p}\beta_k x_{ik}(t)\right\}}.$$

For two individuals at time t,

$$HR_{a,b}(t) = \exp\left\{\sum_k \beta_k[x_{ak}(t) - x_{bk}(t)]\right\}.$$

The coefficients are constant in this representation, but a ratio between two particular individuals may vary because their covariate trajectories vary. Non-informative censoring is assumed conditionally on the available covariate history under the stated model.

For a time-varying path, survival accumulates its **whole history**:

$$S_i(t) = \exp\left[-\int_0^t h_0(u)\exp\{x_i(u)^\top\beta\}\,du\right].$$

In general I cannot simplify this to $S_0(t)^{\exp(x_i^\top\beta)}$, because $x_i(u)$ is not constant as u varies.

Partial likelihood with changing covariates

At each event time $t_{(i)}$, replace fixed x_j by the values at that time:

$$L_p(\beta) = \prod_{i:\delta_i=1}\frac{\exp\{x_i(t_i)^\top\beta\}}{\sum_{j\in R_i}\exp\{x_j(t_i)^\top\beta\}}.$$

Newton–Raphson estimation then proceeds in much the same way. For values not actually measured at event times, possible choices include last observation carried forward, nearest measurement, and interpolation. Those are **assumptions about the longitudinal covariate process**, not innocent computational conveniences: one must avoid using information that would only become available *after* the failure time.

A convenient practical data format is a start–stop record $(t_{\text{start}}, t_{\text{stop}}, \delta, x)$ for each interval during which the recorded covariate values remain fixed.

Bone marrow transplant example

A bone-marrow-transplantation (BMT) study examines disease-free survival and intermediate events during recovery: acute graft-versus-host disease (aGVHD), chronic GVHD (cGVHD), and platelet recovery. It uses two risk-group indicators for ALL and high-risk AML, and defines changing binary event indicators such as

$$z_A(t) = \begin{cases} 0, & t < t_A, \\ 1, & t \geq t_A, \end{cases} \quad z_C(t) = \begin{cases} 0, & t < t_C, \\ 1, & t \geq t_C, \end{cases} \quad z_P(t) = \begin{cases} 0, & t < t_P, \\ 1, & t \geq t_P. \end{cases}$$

The variables must turn on **at the actual intermediate-event time**, not at the baseline. Illustrative models show platelet recovery with a sizable negative coefficient -1.1297 (Wald $p = 0.0006$), whereas the reported aGVHD and cGVHD coefficients are less persuasive in the displayed comparisons. I should not assign a baseline "ever recovered"variable from future knowledge; that would change the definition of the risk sets.

Stratification when a particular factor violates PH

If some grouping variable does not satisfy a common baseline pattern, a **stratified Cox model** gives each stratum $s = 1, \dots, K$ its own baseline hazard:

$$h_s(t \mid x) = h_{0s}(t) \exp(x^\top \beta).$$

The coefficient vector β stays common across strata, and the partial log-likelihood adds the stratum-specific contributions:

$$\ell_p(\beta) = \sum_{s=1}^{K} \ell_{p,s}(\beta).$$

Thus the stratifying factor is allowed to alter the entire baseline hazard pattern, while the remaining covariate effects are constrained to be comparable across strata. The assumption of common coefficients can be checked using Wald or likelihood-ratio comparisons against models with stratum-specific coefficients.

In the BMT example, methotrexate (MTX) prophylaxis motivated separate baseline hazard patterns. A stratified analysis reports of group effects with Wald chi-square 8.916 and $p = 0.0116$. The conceptual difference is what matters: **stratification handles different baseline shapes; a time interaction handles an effect that varies with time**.

⟵ **My thought**

My rule: split records when a person's predictor changes; stratify when a grouping factor needs a different baseline hazard.

Parametric survival distributions and Weibull AFT

Why commit to a survival distribution?

A Cox model avoids choosing a baseline hazard shape. Parametric models assume a probability family for T and can therefore return smooth hazards, survival probabilities, event-time quantiles, and expected lifetimes when those quantities exist. They may be more efficient **if the distribution is correctly chosen**, but can also be more sensitive to model misspecification.

Useful parametric families include exponential, Weibull, log-normal, log-logistic and Pareto models. I want to connect each one through $S(t)$, $H(t) = -\log S(t)$ and $h(t) = -S'(t)/S(t)$ rather than memorize disconnected formulas.

Exponential: constant hazard

For $T \sim \text{Exp}(\lambda)$, $\lambda > 0$,

$$S(t) = e^{-\lambda t}, \quad H(t) = \lambda t, \quad h(t) = \lambda, \quad E(T) = 1/\lambda.$$

This distribution is memoryless and makes the hazard constant. A plot of $\log S(t)$ against t should be roughly linear if the exponential model is plausible; the slope is $-\lambda$.

Weibull: a shape parameter controls changing hazard

A common two-parameter parameterization is

$$S(t) = \exp\left[-\left(\frac{t}{\alpha}\right)^{\gamma}\right], \quad H(t) = \left(\frac{t}{\alpha}\right)^{\gamma}, \quad h(t) = \frac{\gamma t^{\gamma-1}}{\alpha^{\gamma}}, \qquad \alpha, \gamma > 0.$$

For $\gamma < 1$, hazard decreases with time; for $\gamma = 1$ it is exponential/constant; for $\gamma > 1$ it increases. A three-parameter Weibull also includes a location parameter μ:

$$S(t; \mu, \alpha, \gamma) = \exp\left[-\left(\frac{t-\mu}{\alpha}\right)^{\gamma}\right], \quad t > \mu.$$

Here μ represents the lower support boundary. The Weibull family is notable because it can be expressed both as a proportional-hazards regression model and as an accelerated-failure-time model.

A simple distributional check is

$$\log[-\log S(t)] = \gamma \log t - \gamma \log \alpha.$$

A complementary-log-log survival transform plotted against $\log t$ should form an approximately straight line, with slope γ under this scale parameterization.

Why a Gumbel error appears when taking log times

Let $Y = \log T$ for $T \sim \text{Weibull}(\alpha, \gamma)$. Then

$$P(Y > y) = \exp\{-\exp[\gamma(y - \log \alpha)]\}.$$

Writing $\mu = \log \alpha$ and $\sigma = 1/\gamma$ gives

$$\log T = \mu + \sigma\varepsilon,$$

where ε has the standard extreme-value (minimum-type / log-Weibull) survival $P(\varepsilon > z) = e^{-e^{z}}$. This is the bridge from a hazard distribution to a linear **location–scale model for** $\log T$.

Log-normal survival times

If $\log T \sim N(\mu, \sigma^2)$, then

$$S(t) = 1 - \Phi\left(\frac{\log t - \mu}{\sigma}\right), \quad f(t) = \frac{\phi((\log t - \mu)/\sigma)}{\sigma t}, \quad h(t) = \frac{f(t)}{S(t)}.$$

This family is useful for positive, right-skewed survival times; its hazard may rise and then fall instead of being forced to be monotone. A normal-probability transformation of the event-time distribution against $\log t$ is useful for model assessment.

Log-logistic survival times

For scale parameter $\alpha > 0$ in this parameterization and shape $\gamma > 0$,

$$S(t) = \frac{1}{1 + (\alpha t)^{\gamma}}, \qquad h(t) = \frac{\gamma \alpha^{\gamma} t^{\gamma-1}}{1 + (\alpha t)^{\gamma}}.$$

The transformation

$$\log\left\{\frac{S(t)}{1-S(t)}\right\} = -\gamma(\log t + \log\alpha)$$

is linear in $\log t$ under this model. For scale one, the hazard $\gamma t^{\gamma-1}/(1+t^{\gamma})$ decreases if $\gamma \leq 1$. If $\gamma > 1$, it first increases, reaches a maximum at $t = (\gamma-1)^{1/\gamma}$, then decreases. This provides more flexible patterns than Weibull's monotone hazard.

My derivation check: differentiate $-\log S(t) = \log(1+(\alpha t)^{\gamma})$ to recover the stated hazard. Its time derivative then determines the location of the peak.

Pareto and other candidate models

Pareto and generalized Pareto forms are further possibilities. For a Type-I Pareto with $t \geq \sigma$,

$$S(t) = \left(\frac{t}{\sigma}\right)^{-\alpha}, \qquad h(t) = \frac{\alpha}{t}.$$

The $1/t$ hazard is a useful distinguishing feature. A straight line of $\log S(t)$ against $\log t$ is suggested by the formula. I should choose between distributions using both scientific context and adequacy checks, rather than selecting a familiar density by habit.

From log-linear hazard to accelerated failure time

For exponential regression, constant baseline hazard together with a Cox predictor gives

$$h(t \mid x) = \lambda_0 \exp(x^{\top}\beta_{PH}), \qquad \log h(t \mid x) = \log\lambda_0 + x^{\top}\beta_{PH}.$$

For a general AFT model, the regression is instead on log time:

$$\boxed{\log T = \mu + x^{\top}\beta_{AFT} + \sigma\varepsilon}, \qquad T = T_0 \exp(x^{\top}\beta_{AFT}).$$

Consequently $e^{\beta_{AFT,j}}$ is the **time ratio** for a one-unit difference. Positive $\beta_{AFT,j}$ lengthens survival time by that multiplicative factor. The hazard transformation corresponding to any AFT baseline is

$$h(t \mid x) = e^{-x^{\top}\beta_{AFT}}\, h_0\left(te^{-x^{\top}\beta_{AFT}}\right).$$

For Weibull shape γ, substitution shows

$$h(t \mid x) = h_0(t)\exp\{-\gamma x^{\top}\beta_{AFT}\}.$$

So the same Weibull model can be written in PH form with $\beta_{PH} = -\gamma\beta_{AFT}$ under these definitions. Both coefficient conventions appear in Weibull regression; I find it safer to state the exact parameterization before comparing hazard ratios and time ratios.

For a binary treatment predictor with $\beta_{AFT} = 0.4$, the time ratio is $e^{0.4} \approx 1.49$: event times are about 49% longer on the modeled scale than the baseline group. With coefficient -0.4, the time ratio is about 0.67, roughly 33% shorter.

⟵ **My thought**

Big contrast: PH coefficient → hazard ratio; AFT coefficient → time ratio. Weibull permits translating one into the other by its shape parameter.

Parametric regression, AFT interpretation and applications

General AFT model and baseline-time acceleration

Write

$$\log T = \mu + x^\top \beta + \sigma\varepsilon, \qquad T_0 = e^{\mu+\sigma\varepsilon}.$$

Then $T = T_0 e^{x^\top \beta}$, so the model accelerates or slows the event-time clock. A single coefficient β_j shifts log time by β_j per unit of x_j and multiplies all baseline time quantiles by e^{β_j} when the same distributional scale parameter applies. This direct quantile interpretation is one reason I like AFT models.

Log-normal AFT

With $\varepsilon \sim N(0,1)$,

$$\log T = \mu + x^\top \beta + \sigma Z, \qquad Z \sim N(0,1).$$

It follows that

$$S(t \mid x) = 1 - \Phi\left(\frac{\log t - \mu - x^\top \beta}{\sigma}\right).$$

All the covariate effects enter as a shift of the mean of log time, and e^{β_j} is a multiplicative change in the event-time scale. This is not generally a proportional-hazards model; the hazard ratio can vary with time.

Log-logistic AFT and proportional odds

Suppose ε has a standard logistic distribution and $\log T = \mu + x^\top \beta + \sigma\varepsilon$. Then

$$S(t \mid x) = \frac{1}{1 + \exp\{(\log t - \mu - x^\top \beta)/\sigma\}}.$$

Rearranging the survival odds yields

$$\log \frac{S(t \mid x)}{1 - S(t \mid x)} = -\frac{\log t - \mu}{\sigma} + \frac{x^\top \beta}{\sigma}.$$

Thus the log-logistic AFT model also has a **proportional-odds representation for survival**. The effect on log survival odds is β_j/σ under this parameterization. This differs from Cox proportional hazards even though both use an exponential association parameter.

I find it useful to remember which survival families connect to which regression representation: exponential and Weibull can be parameterized as both PH and AFT; log-normal is naturally an AFT model; log-logistic admits AFT and proportional odds.

Choosing a parametric distribution from diagnostic transformations

Compare transformed KM or fitted survival estimates with the linear patterns expected under each candidate family:

Candidate event-time family	Transformation expected to look linear
Exponential	$\log S(t)$ versus t
Weibull	$\log[-\log S(t)]$ versus $\log t$
Log-logistic	$\log\{S(t)/(1-S(t))\}$ versus $\log t$
Log-normal	$\Phi^{-1}\{1-S(t)\}$ versus $\log t$

These checks are empirical diagnostics, not stand-alone proofs of a distribution. In a fitted AFT model, the standardized log-time residual is

$$\hat{\varepsilon}_i = \frac{\log t_i - \hat{\mu} - x_i^\top \hat{\beta}}{\hat{\sigma}}.$$

Its distribution can be compared with the assumed error distribution using a probability plot, taking censoring into account. Cox–Snell and martingale residual ideas can also be adapted when S and h are parametric.

Likelihood under right censoring

For a fully specified density $f(t \mid x; \theta)$ and survival function $S(t \mid x; \theta)$, the observed-data likelihood under non-informative right censoring is, up to the censoring mechanism,

$$\boxed{L(\theta) = \prod_{i=1}^{n} f(t_i \mid x_i; \theta)^{\delta_i} S(t_i \mid x_i; \theta)^{1-\delta_i}}.$$

Equivalently each contribution is $\{h(t_i \mid x_i; \theta)\}^{\delta_i} S(t_i \mid x_i; \theta)$. Here θ contains intercept, coefficient vector, scale and any family-specific shape parameters. Unlike Cox partial likelihood, this approach uses the actual distributional form for event and censoring contributions.

Worked study outline: pharmacotherapy and smoking relapse

The `pharmacoSmoking` example from the R package `asaur` in a study of combination medication versus nicotine patch alone. The key variables are `ttr` (time to relapse), `relapse` (event indicator), `grp` (treatment), and `age` (age at randomization). A useful analysis sequence is:

1. Define relapse time and censoring status, summarize the two treatment groups, and compare groups with KM and a log-rank test.
2. Check whether a stratified log-rank comparison across age groups (≤ 49 versus ≥ 50 in this example) substantively changes the result; the reported difference is small.
3. Fit an initial Cox model for treatment and examine martingale residual patterns against continuous predictors before choosing their functional forms.
4. Examine whether a Weibull event-time distribution is plausible using a complementary-log-log diagnostic.
5. Compare a Cox model with a parametric Weibull-based analysis and explain why hazard-ratio and time-ratio coefficients are not numerically interchangeable.

I focus on the interpretation rather than reproducing executable R listings. R functions/packages associated with this study include `survival::Surv()`, `survfit()`, `survdiff()`, `coxph()`, `survreg()` and the example dataset from `asaur`.

Competing risks and frailty

Why two new ideas are needed

Standard survival analysis is often framed as one type of event. But a patient can die of another cause before the disease outcome of interest; a judge can die or retire; subjects in a clinic can share unmeasured circumstances. Two useful extensions are: **competing risks** for multiple mutually exclusive failure types, and **frailty** for unobserved risk heterogeneity / within-group dependence.

Cause-specific hazard and its relationship to overall hazard

Let $J \in \{1, \dots, m\}$ index the cause of failure. The cause-j hazard is

$$h_j(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T < t + \Delta t,\ J = j \mid T \geq t)}{\Delta t}.$$

When the causes are exhaustive and mutually exclusive,

$$h(t) = \sum_{j=1}^{m} h_j(t), \qquad H_j(t) = \int_0^t h_j(u)\,du, \qquad S(t) = \exp\left\{-\sum_{j=1}^{m} H_j(t)\right\}.$$

For example, "death from cancer"and "death from other causes"contribute to the total instantaneous death risk. Cause-specific hazards condition on being free of **every competing event** until time t.

Cumulative incidence function (CIF)

The object most relevant to "How likely is event type j by time t?"is

$$\boxed{F_j(t) = P(T \le t, J = j) = \int_0^t S(u-)\,h_j(u)\,du}.$$

Consequently,

$$\sum_{j=1}^{m} F_j(t) = 1 - S(t), \qquad \lim_{t\to\infty} F_j(t) = P(J = j)$$

when everyone eventually experiences one of the recorded causes. Another quantity is $P(J = j \mid T \le t) = F_j(t)/(1 - S(t))$, which describes the share of observed failures by time t due to cause j.

Why I cannot just censor other causes and take one minus KM: that approach acts as though competing failures could be removed without changing the relevant probability process. It generally overestimates the actual incidence of cause j. The correct nonparametric estimator uses the probability of surviving *all* causes just before each event time:

$$\widehat{F}_j(t) = \sum_{t_{(i)}\le t} \widehat{S}(t_{(i)}-)\frac{d_{ij}}{r_i},$$

where d_{ij} is the number of events of type j at time $t_{(i)}$ and r_i the all-cause risk-set count. This combines a KM estimate for survival from any cause with an estimate of the type-specific conditional event probability.

⟵ **My thought**

My key contrast: a cause-specific hazard is an instantaneous rate among people still event-free; CIF is an actual probability of that particular event by a time.

Latent failure times and what is observed

An alternative conceptual construction imagines latent times $T_1, \dots, T_m$, one for each potential failure type. Only

$$T = \min_j T_j, \qquad J = \arg\min_j T_j$$

is observed. The joint latent survival is

$$S_L(t_1, \dots, t_m) = P(T_1 > t_1, \dots, T_m > t_m),$$

and actual overall survival is $S(t) = S_L(t, \dots, t)$. The latent joint dependence structure is generally not identifiable from the observed earliest time and cause without further assumptions. That explains why I should interpret the identifiable cause-specific hazards and CIFs directly.

Regression for competing events: two distinct hazard targets

A cause-specific proportional-hazards model for event type j is

$$h_j(t \mid x) = h_{0j}(t) \exp(x^\top \beta_j).$$

It can be fit with a Cox-style partial likelihood by treating failures of cause j as events and other causes as censored **for the cause-specific hazard estimand**. This is a mathematical way to estimate the instantaneous cause-specific rate; it does not mean that other events are ordinary non-informative censoring for a cumulative-incidence probability. An alternative model targets a **subdistribution hazard** associated with $F_j(t)$, using a different risk-set construction (often called Fine–Gray regression). The distinction matters: cause-specific and subdistribution hazard ratios answer different scientific questions.

An illustrative example compares death and retirement from the U.S. Supreme Court. The reported illustrative hazard ratios per additional year of age are about 1.084 for departure by any cause, 1.071 for death and 1.106 for retirement; calendar-year factors are reported as 0.994, 0.989 and 0.999, respectively. The exercise is to interpret the *cause-specific* hazards before trying to convert them into cumulative probabilities.

Frailty as a random effect in survival analysis

Even after including observed predictors, two individuals may differ in unmeasured vulnerability. In a shared-frailty model for individual/cluster i and episode/member j,

$$h_{ij}(t \mid u_i, x_{ij}) = u_i h_0(t) \exp(x_{ij}^\top \beta), \qquad u_i > 0.$$

The positive random effect u_i is the **frailty**. An equivalent log-risk model is

$$h_{ij}(t) = h_0(t) \exp\{x_{ij}^\top \beta + \sigma \omega_i\}, \qquad u_i = e^{\sigma \omega_i}.$$

If frailty variance tends to zero, the extra heterogeneity disappears and we recover the corresponding standard Cox model. Shared frailty means members of the same family, clinic or cluster have a common unmeasured effect, inducing dependence between their event times. Individual frailty assigns a separate unobserved effect to each person, often to model heterogeneity rather than clustering.

Conditional independence and marginal dependence

For two event times sharing u, conditional independence gives

$$S_{12}(t_1, t_2 \mid u) = \exp\{-u[H_{01}(t_1) + H_{02}(t_2)]\}.$$

Averaging over the frailty distribution produces a marginal joint survival function:

$$S_{12}(t_1, t_2) = E_u\left[e^{-u(H_{01}(t_1)+H_{02}(t_2))}\right].$$

For Gamma frailty with mean 1 and variance $\theta > 0$, the Laplace transform yields the explicit form

$$\boxed{S_{12}(t_1, t_2) = \{1 + \theta[H_{01}(t_1) + H_{02}(t_2)]\}^{-1/\theta}}.$$

I particularly like this example because a distributional transform—the Laplace transform of the random effect—turns unobserved heterogeneity into a tractable survival model. Other choices include inverse-Gaussian frailty and nonparametric finite mixtures with weights π_k and values u_k:

$$S_{12}(t_1, t_2) = \sum_{k=1}^{K} \pi_k \exp\{-u_k[H_{01}(t_1) + H_{02}(t_2)]\}.$$

The main concept is **conditional independence does not imply marginal independence after integrating out a shared random effect**.

Software and analysis cautions

In R, relevant tools include `survival::coxph()` with a frailty term and the `coxme` package / `coxme()` for mixed-effects Cox models. For competing risks, cause-specific Cox regressions and CIF estimation address different targets; their outputs should be labeled accordingly. None of these tools can rescue a seriously misspecified event definition or an incorrect treatment of censoring and competing events.

My final connections and quick self-test

The course now forms a useful chain in my mind:

1. **Cox PH:** compare instantaneous risk while leaving baseline hazard unspecified. Partial likelihood works because the baseline cancels from risk-set comparisons.
2. **Model assessment:** likelihood-ratio tests compare nested specifications; a c-index measures ranking; a time-dependent ROC measures discrimination at a defined horizon; residuals probe distinct assumptions.
3. **Time dependence:** a changing predictor and a changing coefficient are different models. The former updates the subject state; the latter relaxes proportional hazards.
4. **Parametric models / AFT:** assuming a distribution permits direct time-scale or quantile predictions. Weibull uniquely connects PH and AFT among the main families studied here.
5. **Competing risks:** there is a difference between the hazard of a particular cause and the actual probability that it occurs before another cause.
6. **Frailty:** shared unobserved factors explain why observed event times can remain dependent after adjusting for measured covariates.

Questions I want to be able to answer without looking back:

- Derive the partial likelihood for one untied event and show exactly which factor cancels.
- Explain why $\exp(\beta)$ is a hazard ratio in a Cox model but a time ratio in an AFT model.
- Compute a fitted baseline survival curve from a Cox coefficient and a baseline cumulative hazard estimate.
- Distinguish Cox–Snell, martingale and Schoenfeld residuals by their definitions and goals.
- Explain when $x(t)$ versus $\beta(t)$ is appropriate and why current covariate values belong in the risk set.
- Derive $F_j(t) = \int_0^t S(u-)h_j(u)du$ and explain why censoring other causes in KM is generally wrong for a CIF.
- Derive the Gamma frailty joint survival using $E(e^{-uz})$ and explain shared dependence.